

EX NO: 9	SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF ENCODED OUTPUT BIT SEQUENCES USING MATLAB
DATE:	

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Fixed Input Sequence
```

```
input_bits = [1 0 0 1 1];
```

```
%% Convolutional Encoder
```

```
trellis = poly2trellis(3, [7 5]);
```

```
encoded_bits = convenc(input_bits, trellis);
```

```
%% Separate encoded bits into pairs
```

```
encoded_pairs = reshape(encoded_bits, 2, []);
```

```
%% Create output strings
```

```
output_pair = cell(length(input_bits), 1);
```

```
for i = 1:length(input_bits)
```

```
    output_pair{i} = sprintf('%d%d', ...
```

```
        encoded_pairs(i,1), encoded_pairs(i,2));
```

```
end
```

```

%% Create table

Bit_Position = (1:length(input_bits));

Input_Bit = input_bits;

Output_Encoded = output_pair; % FIXED

T = table(Bit_Position, Input_Bit, Output_Encoded);

disp('Convolutional Encoding Table');
disp(T);

%% Plot input and encoded bit sequences

figure;

subplot(2,1,1);
stairs(input_bits, 'LineWidth', 1.5);
ylim([-0.2 1.2]);
title('Input Bit Sequence');
xlabel('Bit Position');
ylabel('Bit Value');
grid on;

subplot(2,1,2);
stairs(encoded_bits, 'LineWidth', 1.5);
ylim([-0.2 1.2]);
title('Encoded Output Bit Sequence');
xlabel('Encoded Bit Position');
ylabel('Bit Value');

```

grid on;

Convolutional Encoding Table		
Bit_Position	Input_Bit	Output_Encoded
1	1	{ '11' }
2	0	{ '10' }
3	0	{ '11' }
4	1	{ '11' }
5	1	{ '01' }

